# DMDW6 — DATASET LINKS, DATA & HOW TO USE

*Apriori / Association Rule Mining — Quick Reference*

## Dataset shortlist

| Dataset | Domain | What data is inside? | How to use with Apriori | Link |
|---|---|---|---|---|
| Disease Symptoms and Patient Profile | Healthcare | Patient/disease records containing symptoms and patient-profile information. | Convert symptoms into items such as Fever=Yes, Cough=Yes, Headache=Yes. Use Apriori to find frequent symptom combinations and rules such as symptom combination → disease.<br><br>Preparation: Clean missing values; encode categorical/binary fields as items. Keep disease as a consequent when doing disease-oriented rules. | Open dataset |
| UCI Car Evaluation | Automobiles | 1,728 car records with categorical features: buying price, maintenance price, doors, persons, luggage boot, safety and evaluation class. | Find combinations of car features associated with an acceptable/unacceptable evaluation.<br><br>Preparation: Almost no binning is needed because the attributes are categorical. Convert each row into items such as safety=high, buying=vhigh, persons=4. | Open dataset |
| UCI Mushroom | Biology / Food Safety | 8,124 mushroom records with 22 categorical attributes and an edible/poisonous class. | Discover combinations of mushroom characteristics associated with edible or poisonous classes. | Open dataset |

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| | | | Preparation: Convert every attribute=value into an item. Handle missing stalk-root values carefully. | |
| UCI Soybean (Large) | Plant Disease | 307 plant records with 35 categorical attributes and multiple disease/injury classes. | Discover symptom/condition combinations associated with different plant diseases.<br><br>Preparation: Encode categorical observations as items and handle '?' values appropriately. Filter very rare classes/rules. | Open dataset |
| UCI Automobile | Automobiles | 205 automobile records containing specifications such as make, body style, fuel type, engine size, horsepower, MPG and price. | Find associations among automobile specifications and price/body-style categories.<br><br>Preparation: Categorical columns become items directly. Numerical columns such as price and horsepower must first be binned into low/medium/high or meaningful ranges. | Open dataset |
| Groceries | Retail / Market Basket | Grocery purchase transactions where each basket contains multiple purchased products. | Classic Apriori: discover products frequently purchased together, e.g. bread → milk.<br><br>Preparation: Keep each customer's basket as one transaction. Do not treat each individual product row as a separate transaction. | Open dataset |
| COVID-19 Clinical Data Repository | Healthcare | Clinical records containing | Find co-occurring symptoms/comorbi | Open dataset |

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| | | symptoms, comorbidities and other clinical/demographic variables. | dities and associations with selected clinical outcomes.<br><br>Preparation: Convert binary/categorical clinical fields into items. Clearly document preprocessing and avoid interpreting rules as medical causation. | |

## Basic Apriori workflow

1. Download the dataset from the link.
2. Inspect columns, missing values and categorical/numerical fields.
3. Convert categorical values into items: e.g. safety=high, fuel=gas.
4. For numerical columns, create bins such as price=low/medium/high before Apriori.
5. Create one transaction per row/entity, unless the dataset is already a basket dataset.
6. Run Apriori with minimum support.
7. Generate association rules and filter by confidence and lift.
8. Show the strongest meaningful rules in the project report and explain that association does not imply causation.

## Which dataset requires the least preprocessing?

UCI Car Evaluation and Groceries are particularly straightforward. Car Evaluation is almost entirely categorical, while Groceries is already in transaction/basket form. For a healthcare project, the Disease Symptoms and Patient Profile dataset gives a direct symptom-association story but requires more careful preprocessing.

## Useful rule metrics

- Support: how frequently X and Y occur together in the whole dataset.
- Confidence: how often Y occurs when X occurs.
- Lift: compares the observed X→Y co-occurrence with what would be expected if X and Y were independent. Lift > 1 indicates positive association.

*Prepared as a quick dataset-selection sheet for the DMDW6 project.*